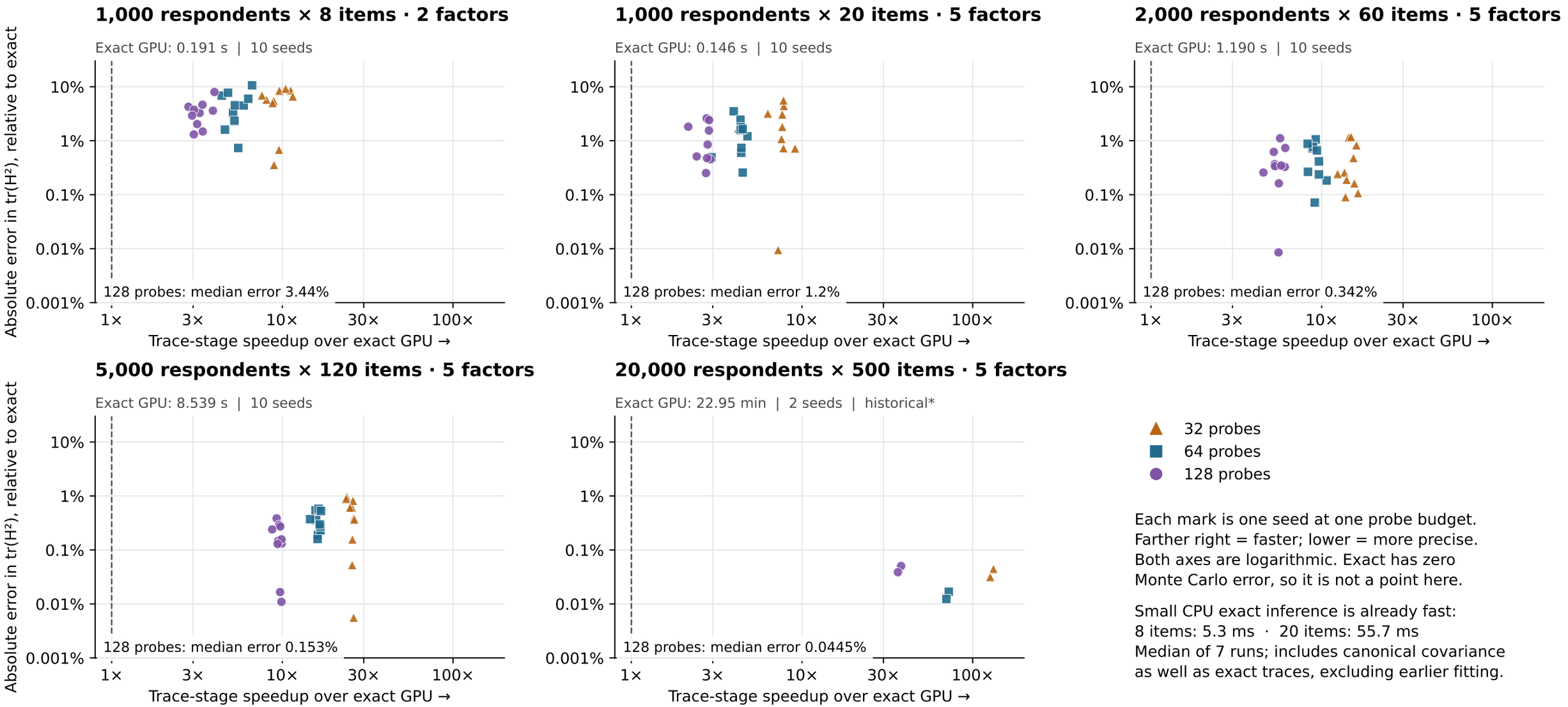
# MV correction: measured speed–precision tradeoff

Both modes apply the correction. Random probes estimate its second trace moment; the DWLS fit and standard errors are unchanged.




Each mark is one seed at one probe budget.
Farther right = faster; lower = more precise.
Both axes are logarithmic. Exact has zero Monte Carlo error, so it is not a point here.

Small CPU exact inference is already fast:
8 items: 5.3 ms · 20 items: 55.7 ms
Median of 7 runs; includes canonical covariance as well as exact traces, excluding earlier fitting.

GPU times include observed kernel startup + trace computation; shared projections/covariance, numerical self-checks and rotations are excluded.
*500-item case reuses the completed exact benchmark and two probe seeds; other cases have fresh exact timings and ten seeds.
Apple M1 · 8 GiB · Chrome 156 · synthetic data. Nested probe checkpoints share random streams. These cases do not establish a universal size threshold.